Mofftasim Hossain Sayem

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Summary

Computer Science graduate focused on backend and full-stack development in Go, TypeScript, and Python. Shipped an open-source app, built LLM features with RAG and tool calling, and reached the finals of two national hackathons in 2026.

Education

BRAC University Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering *2022 – 2026*
- **CGPA:** 3.67/4.00. Final-year thesis grade: **A**+.

Dhaka College | *HSC, Dhaka Board, GPA 5.00/5.00, Merit Scholarship* 2021

Motijheel Ideal School and College | *SSC, Dhaka Board, GPA 5.00/5.00, Merit Scholarship* 2019

Projects

SortWise | *Go, React, TypeScript, SQLite, Gemini API* | GitHub Sep 2026 – Present
- Shipped an open-source app that summarizes, tags, and categorizes X bookmarks with multimodal LLMs.
- Built hybrid search over vector embeddings and SQLite FTS5 to find posts by meaning or keyword.
- Engineered a Go REST API with durable job queues, background workers, and failover from Gemini to Groq.

Polaris | *Next.js 15, TypeScript, MongoDB, Gemma 4* | Live Demo Jul 2026 – Sep 2026
- Built a Bengali-first study planner that turns goals, constraints, and deadlines into weekly roadmaps.
- Built a Gemma 4 agent with typed tool calling, hybrid BM25 and vector RAG, and persistent memory.
- Made batched LLM question generation retry-safe with atomic MongoDB claims and deduplication.

ByteLearn | *PHP, Laravel 10, MySQL* | GitHub Oct 2025 – Dec 2025
- Designed the 17-migration MySQL schema and Laravel backend of a course platform.
- Added session authentication, role middleware, AI-generated quizzes, and a lesson-based chatbot.

WattsUp and Vantage Arm Digital Twin | *Node.js, Socket.IO, React, Three.js* | Live Demo Jul 2026
Techathon Nationals 2026 builds: the live React dashboard of an office power monitor fed by Socket.IO, and a browser digital twin of a 6-DOF robotic arm in Three.js.

Systems Programming | *C, Linux, Flex, Bison* Jan 2025 – Aug 2026
- Wrote a Unix shell with signal handling, a file-system consistency checker (fsck), and a compiler front end.

Research and AI Projects

Obscrowd: Multi-Portrait Privacy under Generative Editing | *PyTorch, Diffusers* Oct 2025 – Jun 2026
Completed a four-member BRAC University thesis (grade A+) developing Obscrowd, a multi-portrait biometric unlinkability framework for protecting faces after generative editing; contributed to the model pipeline and evaluation, and the manuscript is being prepared for publication.

Multi-Modal Music Clustering | *PyTorch, librosa, scikit-learn* Dec 2025 – Jan 2026
- Learned joint audio and lyric embeddings for 450 Bangla and English songs with VAE and Beta-VAE models.
- Benchmarked latent clusters against PCA with K-Means and spectral clustering on silhouette, NMI, and purity.

Hate Speech Detection | *scikit-learn, PyTorch, Hugging Face Transformers* Jan 2025 – May 2025
- Fine-tuned BERT to 90% accuracy on hate speech classification, against SVM and LSTM baselines.

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, C, SQL
Backend and Databases: Node.js, Express, Next.js, REST APIs, Socket.IO, MongoDB, SQLite, MySQL, Docker
Frontend: React, HTML, CSS, Tailwind CSS, Three.js
Machine Learning and AI: PyTorch, Hugging Face Transformers, Diffusers, scikit-learn, OpenCV, RAG
Robotics and Tools: Arduino, I2C sensors, motor and servo control, Git, GitHub, Linux

Achievements

- **Finalist**, Infinity AI BuildFest 2026, a national AI hackathon with 3,500+ registrants (Polaris).
- **Special Recognition (Top Projects)**, Build with Gemma AI Innovation Challenge 2026 (Polaris).
- **National Finalist**, Techathon Nationals 2026, IUT Robotics Society.